



COPPERBELT UNIVERSITY
SCHOOL OF ENGINEERING
MECHANICAL DEPARTMENT
SCHOOL OF ENGINEERING

PROJECT PROPOSAL PRESENTATION

PROJECT TITLE: IMPROVEMENT OF A MOTORIZED
GYROSCOPE (TO MAKE IT OPERATE NON STOP FOR 24 HOURS)

STUDENT DETAILS:

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SUPERVISOR: Mr. Bennet Siyingwa



**AIMS AND
OBJECTIVES**

AIM:

To enhance, evaluate, and sustain a cost-effective motorized gyroscope capable of continuous operation for a minimum of 24 hours.

KEY OBJECTIVES:

1. Develop a dynamically balanced motorized gyroscope capable of stable operation above 5000 RPM for 24 hours.
2. Utilize homogeneous materials to eliminate flywheel imbalances.
3. Build real-time monitoring systems for critical operational parameters.
4. Integrate and optimize components from previous design iterations.



DESIGN CONSIDERATIONS

1. Consider and improve the previous design.

- ✓ This provides a baseline from which to build, learning from the challenges our predecessors encountered and how they overcame them.
- ✓ We save time as we take advantages of the things they did right.
- ✓ Following their simple design simplifies our work allowing us to focus on making the gyroscope run 24+ hours.

2. Friction, High moment of inertial & Angular Moment

- ✓ Friction goes up, efficiency & runtime goes down. (heat, efficiency, wear, life, bearings, noise)
- ✓ High moment of inertia: Reduces the rate of spin decay, we get more stored energy and more stable system.

3. Availability of materials

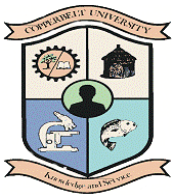
- ✓ Locally accessible metals that are easy to machine into various components.

4. Homogeneous flywheel material

- ✓ Reduces vibrations, energy losses (heat, noise), components breakdown, increases efficiency
- ✓ Dynamically balanced system

WHY THIS DESIGN?:

Simple, Effective And Efficient (Lean Manufacturing), Online Community Support, Saves Time and is Cost Effective.



DESIGN
CALCULATIONS

Data from CAD software and datasheet
m=0.8606kg
ωs=837.76rad/s
ωp=0.5rad/s
I=10A
Motor KV rating 2200KV

FORMULA	APPROXIMATE CALCULATIONS
Moment of Inertia $I = \frac{1}{2}mr^2$ <i>Hibbeler (2015, p.388)</i>	$I = \frac{1}{2} \times 0.8606 \times (0.1079)^2 = 0.005006 \text{ kg}\cdot\text{m}^2$
Gyroscopic Torque $T_g = I\omega_s\omega_p$ <i>Meriam, Kraige and Bolton (2016, p.486)</i>	$T_g = 0.005006 \times 837.76 \times 0.5 = 2.097 \text{ N}\cdot\text{m}$
Angular Momentum $L = I\omega_s$ <i>Hibbeler (2015, p.412)</i>	$L = 0.005006 \times 837.76 = 4.194 \text{ kg}\cdot\text{m}^2/\text{s}$
Stability Criterion $I\omega_s\omega_p \geq mgh\sin(\theta)$ <i>Ogata (2010, p.74)</i>	$2.097 \geq 0.8606 \times 9.81 \times 0.05 \times \sin(5^\circ)$ $2.097 \geq 0.037; \text{ Stable } \checkmark$
Back-EMF Constant $k_e \approx 1/(KV \times 2\pi/60)$ <i>Ogata (2010, p.52)</i>	$k_e = 1 / (2200 \times 0.10472) = 0.00434 \text{ V}\cdot\text{s}/\text{rad}$
Voltage Supplied $V = (\text{RPM}/KV) + (I_m \times R)$ <i>Ogata (2010, p.59)</i>	$V = (8000/2200) + (10.0 \times 0.12) = 4.84 \text{ V}$
Stored Kinetic Energy $E_s = \frac{1}{2}I\omega^2$ <i>Meriam, Kraige and Bolton (2016, p.312)</i>	$E_s = \frac{1}{2} \times 0.005006 \times (837.76)^2 = 1756.7 \text{ J} = 1.757 \text{ kJ}$
Power $P = V \times I = T \times \omega$ <i>Hibbeler (2015, p.201)</i>	Mechanical: $2.097 \times 837.76 = 1756.7 \text{ W}$ Electrical: $4.84 \times 10.0 = 48.4 \text{ W}$

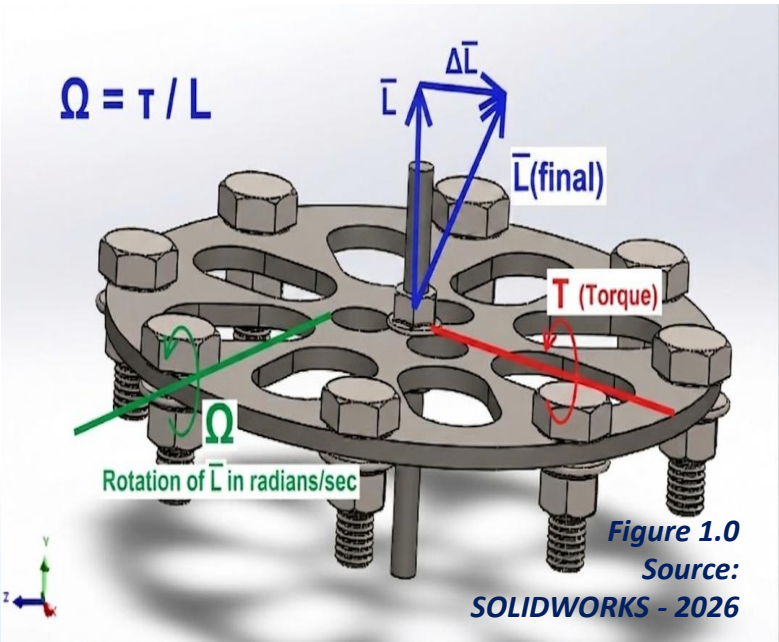


Figure 1.0
Source:
SOLIDWORKS - 2026

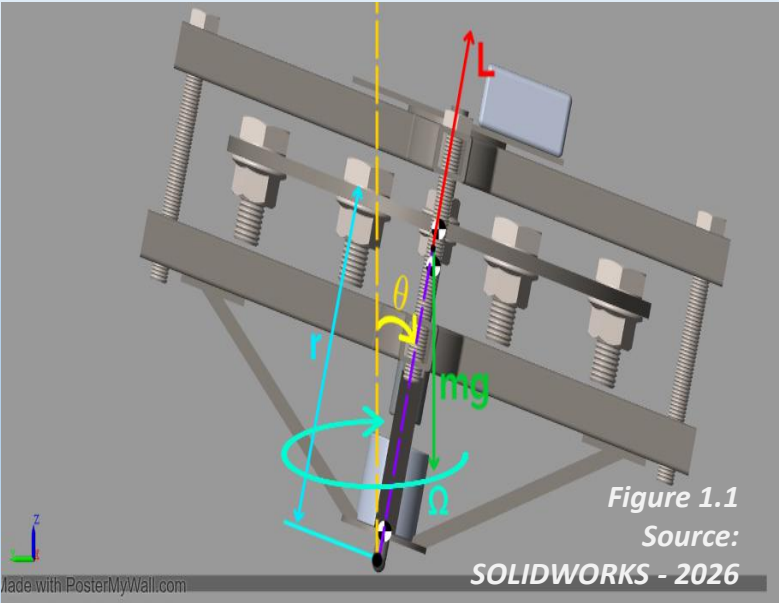
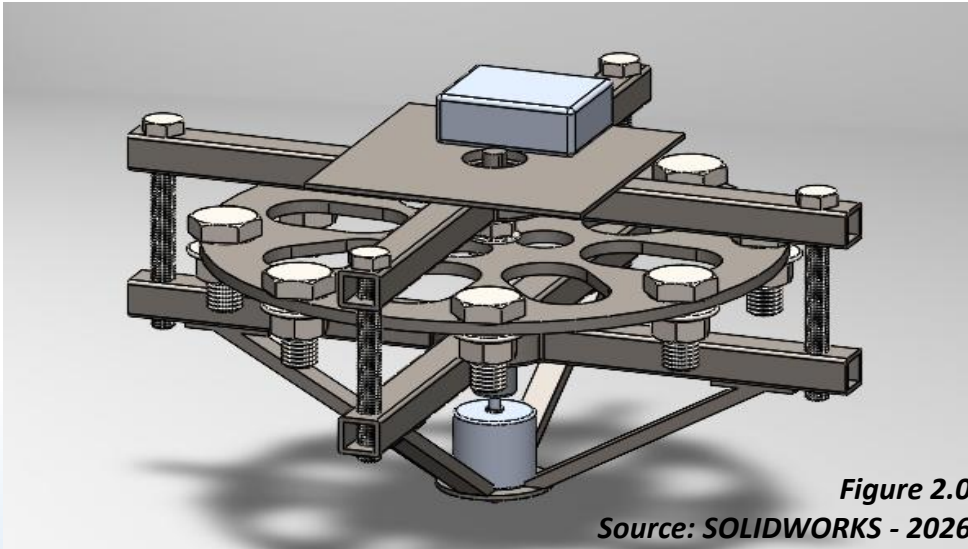


Figure 1.1
Source:
SOLIDWORKS - 2026

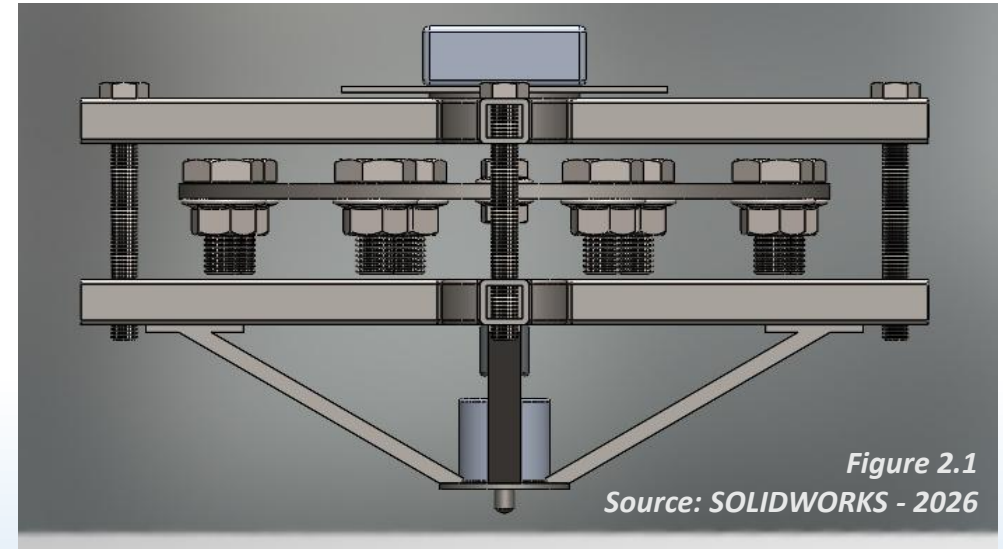


CONCEPTUAL DESIGN

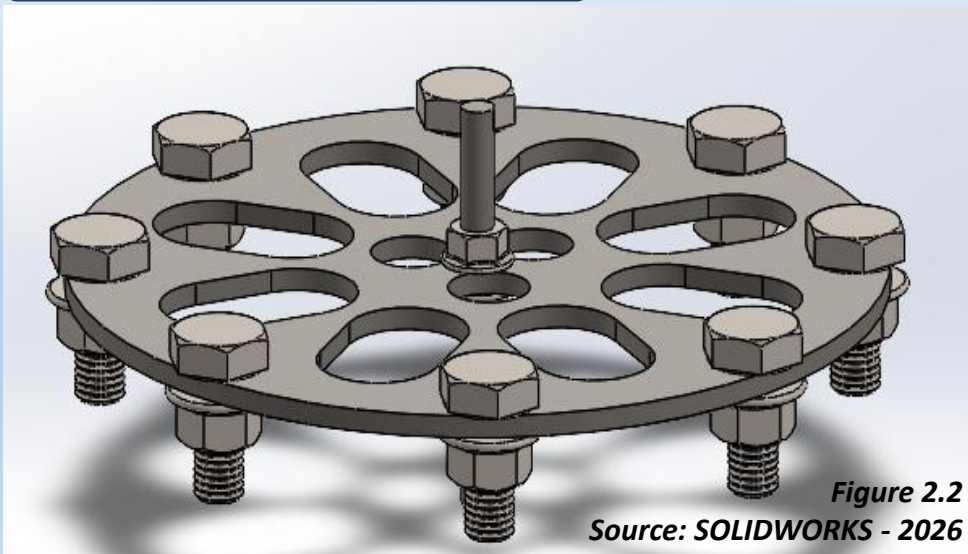
Diametric View



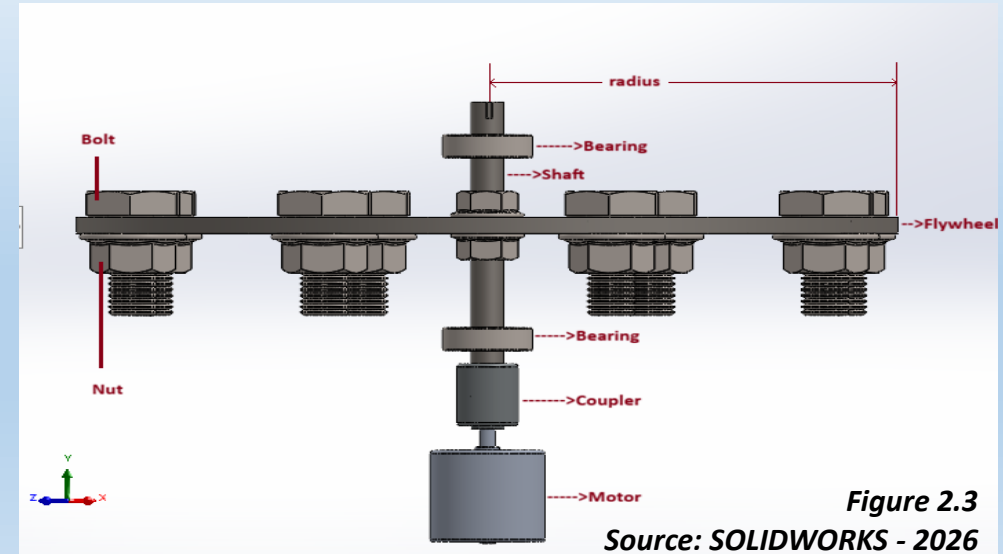
Side View

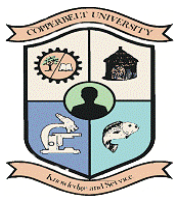


Flywheel Diametric View



Inside Components - Side View





CONCEPTUAL DESIGN

SUBSYSTEMS - MATLAB Simulink

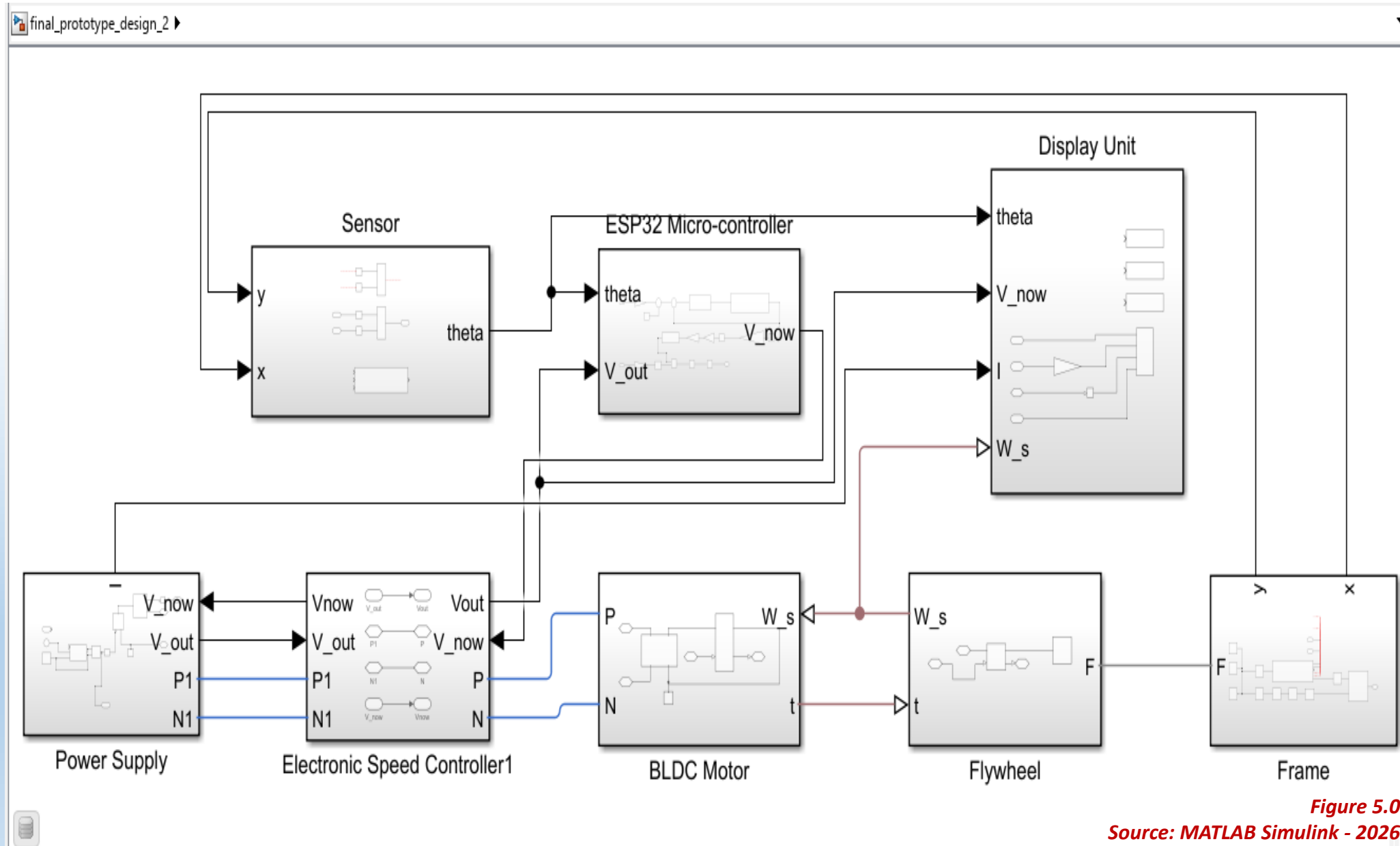


Figure 5.0

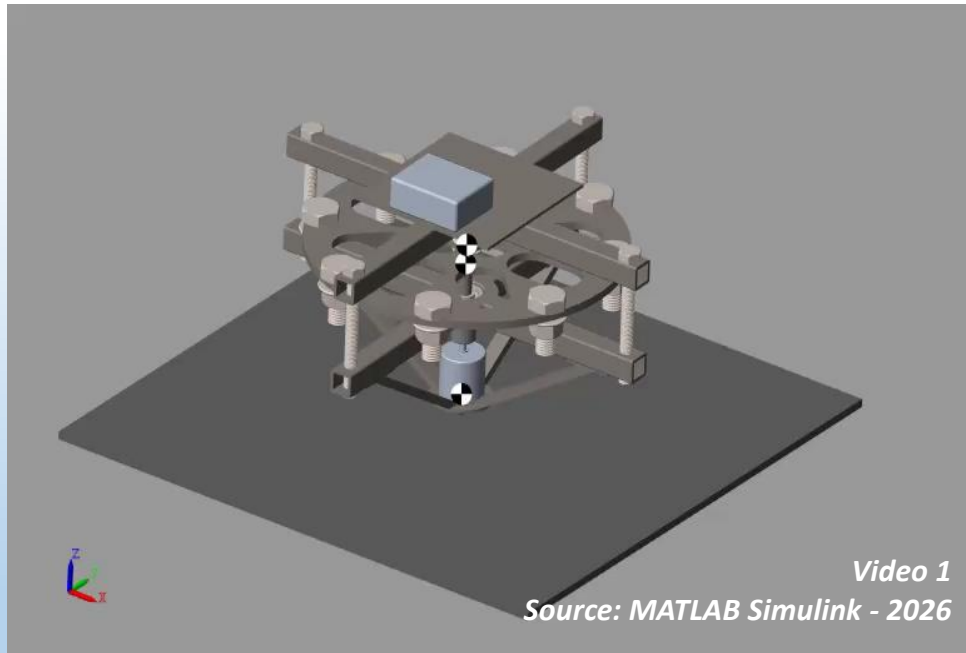
Source: MATLAB Simulink - 2026



DESIGN
SIMULATIONS

MATLAB Simulink Simulations – RESULTS VIDEOS

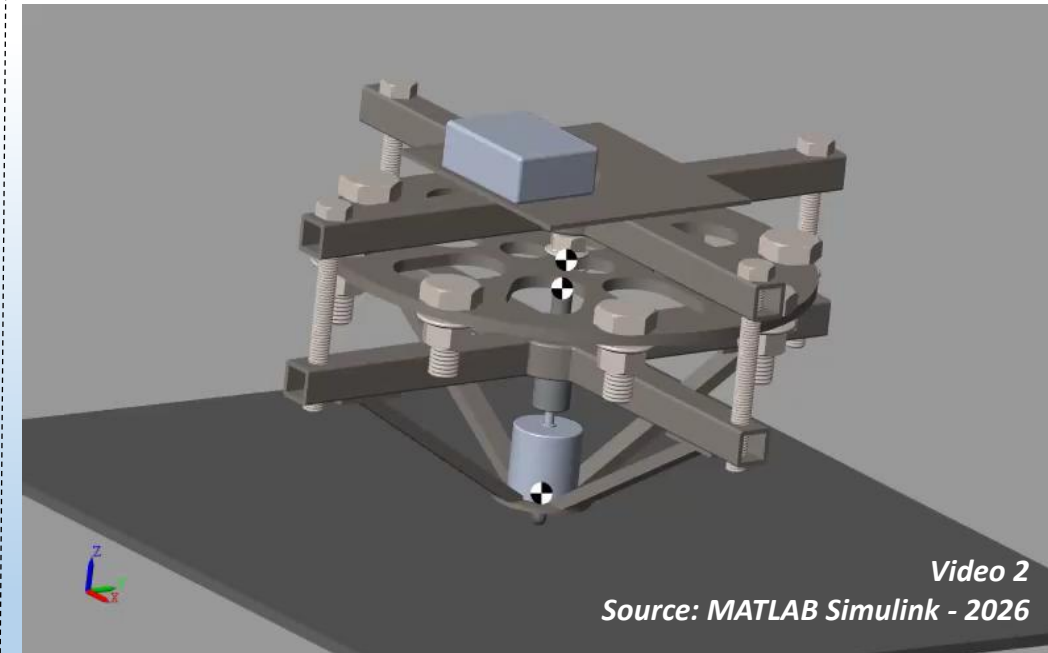
BEFORE CONTROL



INPUT PARAMETERS:

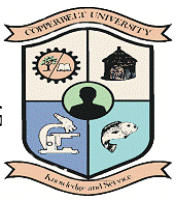
1. Tilt angle (theta): *Initial disturbance factor.*
2. Motor Supply Voltage: *Primary control input.*

CONTROLLED



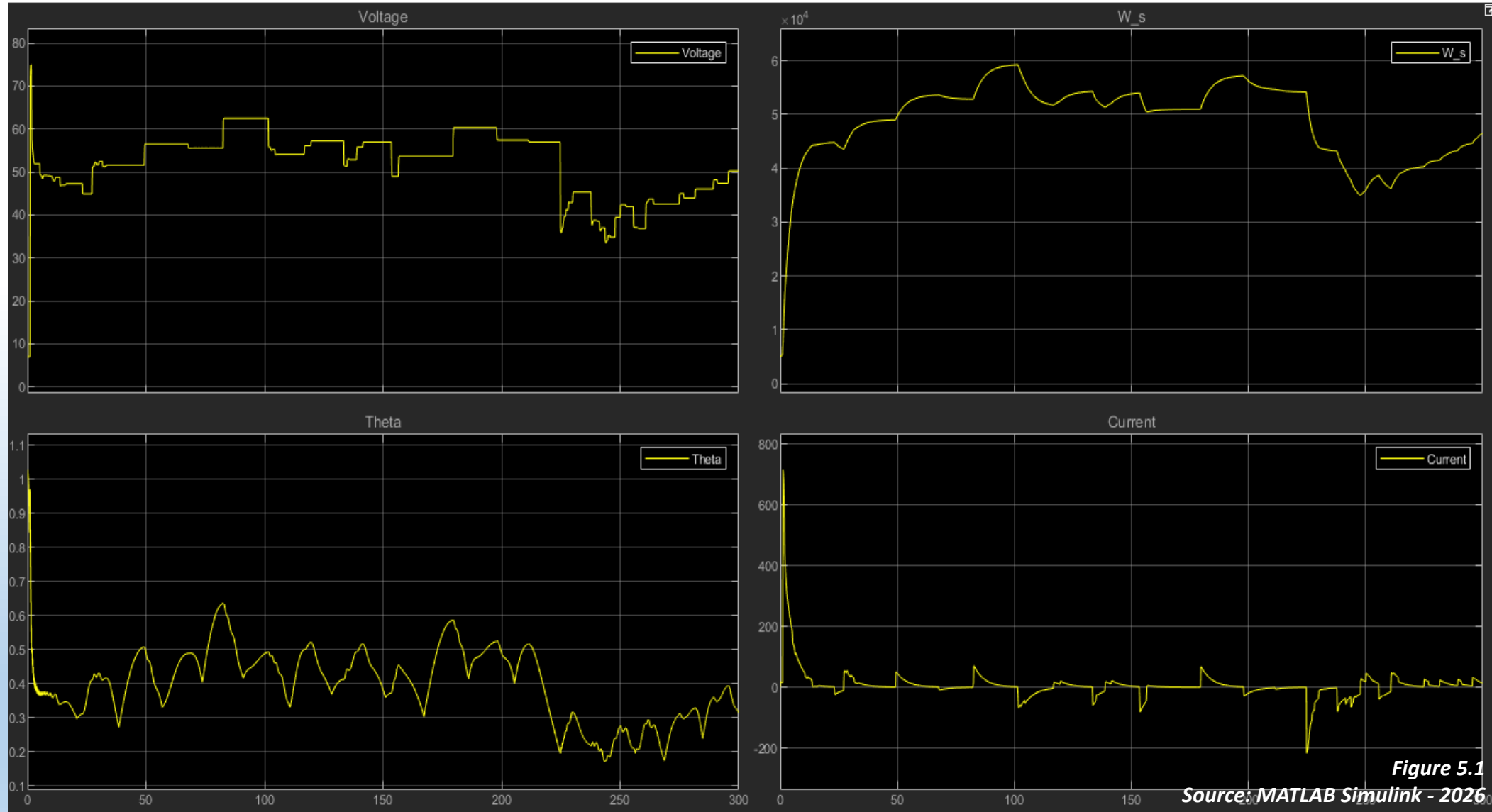
OUTPUT PARAMETERS:

1. Flywheel Rotational Speed
2. Dynamic Tilt Response: *Equilibrium maintenance.*



DESIGN
SIMULATIONS

MATLAB Simulink Simulations – RESULTS (CONTROLLED SYSTEM)

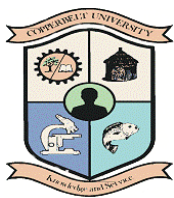


INPUT PARAMETERS:

1. Tilt angle (theta): *Initial disturbance factor.*
2. Motor Supply Voltage: *Primary control input.*

OUTPUT PARAMETERS:

1. Flywheel Rotational Speed
2. Dynamic Tilt Response: *Equilibrium maintenance.*



DESIGN
SIMULATIONS

CIRCUIT DESIGN – PROTEUS Software

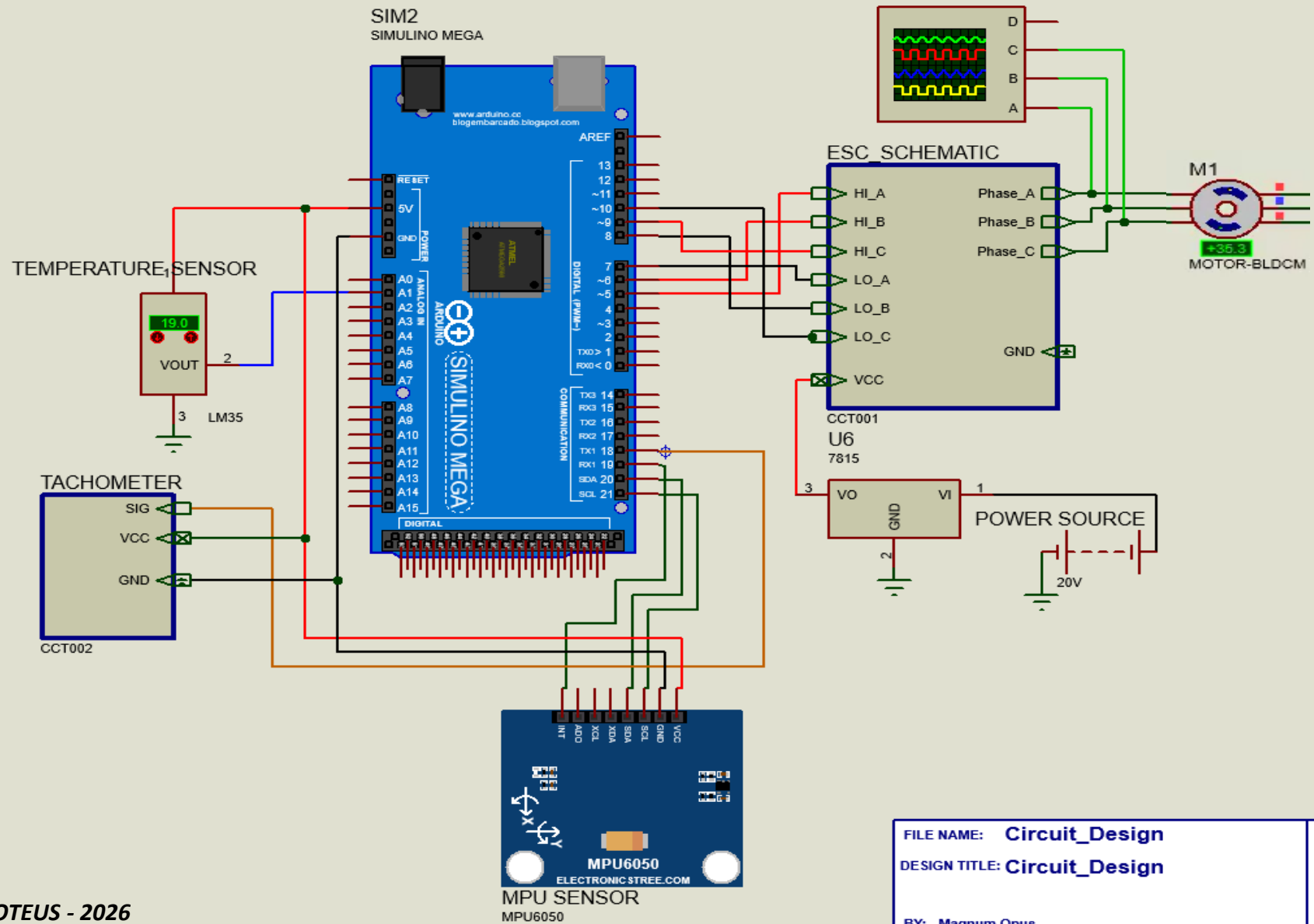
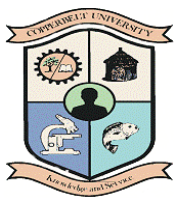


Figure 6.0
Source: PROTEUS - 2026

FILE NAME: Circuit_Design	DATE: 5/10/2026
DESIGN TITLE: Circuit_Design	
BY: Magnum Opus	



MATERIAL
SELECTION

<i>Component / Material</i>	<i>Used in</i>	<i>Specification</i>	<i>Why</i>
Mild Steel	Structural Frame & Flywheel disk	High tensile strength	Easy to work with, locally available and for torque resistance
Steel Rod	Main Shaft	M8 (8mm diameter) precision rod	High Rigidity, Torsional Strength and is Fatigue Resistance.
Steel Bolts & Nuts	Flywheel Assembly	M12 (12mm) and M6 (6mm) variants	To concentrate the mass flywheel to the edges of the disk and also increase its moment of inertia.
Rubber Coupling	Power Transmission	Vibration dampening and alignment	Shock Absorption, Vibration Damping, and Misalignment Compensation.
RS608 Ball Bearings	Rotation Support	Low friction, high-speed durability	Minimizes frictional heat and provides radial support

Table 1

** Materials selected based on local accessibility and suitability for precision CNC machining.*



FLYWHEEL FABRICATION

- ❑ **CNC TURNING:** Precision machining from solid 5mm mild steel plate to ensure geometric accuracy.
 - **DYNAMIC BALANCING:** Post-process rig testing to identify and drill out heavy spots, minimizing vibration.

STRUCTURAL ASSEMBLY

- ❑ **MAIN FRAME:** Robust construction using high-strength welding and secure bolt-nut fastening.
- ❑ **SHAFT INTEGRATION:** Interference fit and set screws for reliable power transmission.

ELECTRONIC INTEGRATION

- ❑ **ROBUST CIRCUITRY:** Transitioning from breadboards to soldered Perf-boards for mechanical stability.
 - **SIGNAL INTEGRITY:** Ensuring stable I2C communication between ESP32 and MPU6050 under high-vibration.

Design Standard: To minimize eccentricity and parasitic torque, 0.5mm machining tolerance maintained.

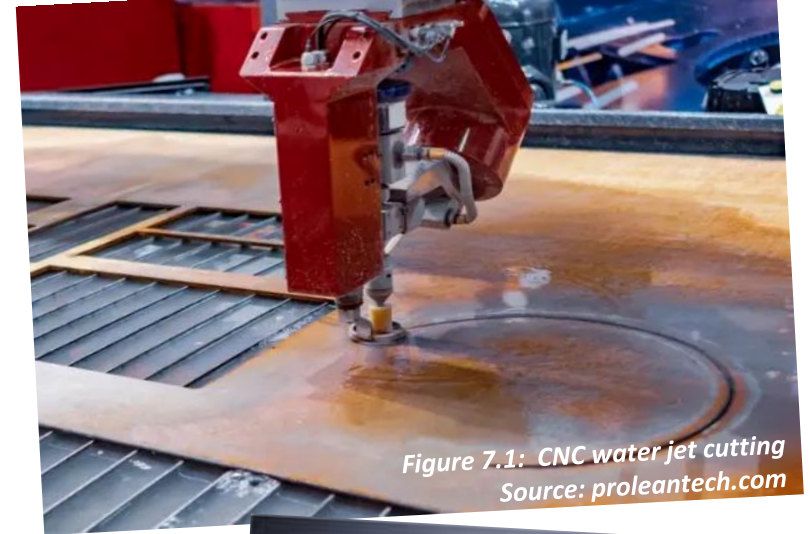


Figure 7.1: CNC water jet cutting
Source: proleantech.com



Figure 7.2
Source: aeron.co.uk

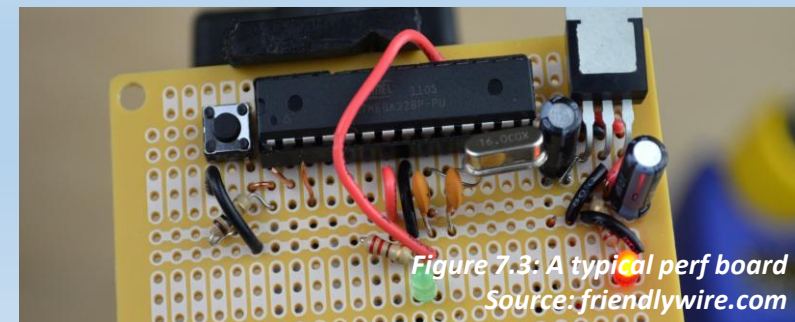


Figure 7.3: A typical perf board
Source: friendlywire.com



REMAINING
WORK

1. COMPONENT ACQUISITION

Procurement and gathering of all mechanical components, electronic sensors, and control hardware.

2. PROTOTYPE CONSTRUCTION

Physical assembly of the gyroscope frame, flywheel flywheel machining, and integration of the drive drive system.

3. OPERATIONAL VALIDATION

Rigorous stress testing to ensure continuous operation exceeding the 24-hour benchmark.

4. FINAL PRESENTATION

Documentation of results and final project defense before the evaluation panel.



*Figure 8.0: BLDC Motor
Source: ebay.com*



*Figure 8.1: Electronic
speed controller (ESC)
Source: ebay.com*



CHALLENGES AND MITIGATIONS

Low Motor Starting Torque

Solution: We implemented manual starting of flywheel rotation followed by motor handover to sustain high-speed operation.

SIMULATION LIBRARY LIMITATIONS (ESP32/ESC)

We could not find the right libraries to use in the simulation for ESP32, Electronic speed controller (ESC) so we decided to use Arduino and built our own ESC module. This costed as a lot of time.

MECHANICAL DESIGN CONSTRAINTS

Since this is a continuation project, we followed our predecessors' design. However, it is limited—it only allows one degree of control (changing motor speed), which makes our work extremely difficult.

SCARCITY OF SPECIFIC CONTROL RESOURCES

Despite a gyroscope being a common concept in the online tech world, this particular design is mostly purely mechanical and there is little help on how to make control algorithms and control it. But with the help of AI and MATLAB Simulink, we managed to create some control logics that worked in the simulations.



REFERENCES

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THE END